$$= \left(\frac{11+1}{2}\right)^{th} \text{term}$$

$$= 6^{\text{th}} \text{ term} = 912$$

Therefore, the median is 912.

Hence, the median lifetime of the LED bulb is 912 days.

Example 5.14 Find the Median of the following data.

12, 7, 23, 14, 19, 10, 5, 26

Solution

Arranging the data in ascending order, we have

5, 7, 10, 12, 14, 19, 23, 26.

Here, $n = 8$, which is even.

Therefore, Median $= \frac{1}{2}\left\{\left(\frac{8}{2}\right)^{th} term + \left(\frac{8}{2}+1\right)^{th} term\right\}$

$$= \frac{1}{2}\left\{4^{\text{th}}\ term + 5^{\text{th}}\ term\right\}$$

$$= \frac{1}{2}\{12+14\}$$

$$= \frac{26}{2} = 13$$

Therefore, the Median is 13.

Think

Complete the table given below and observe it to answer the following questions.

Series	Values	Mean	Median
A	99,100,101		
B	90,100,110		
C	50,100,150		
D	99,100,200		

(i) Which are all the series having common mean and median?

(ii) Why median is same for all the 4 series?

(iii) How mean is unchanged in the series A, B and C.

(iv) What change is to be made in the data, so that mean and median of 'D' series is equal to other series?

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